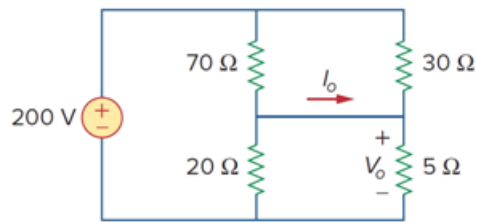


Problem

Calculate V_o and I_o in the circuit of Fig. 2.99.

Figure 2.99:



Step-by-step solution

Step 1 of 5

Refer to Figure 2.99 in the textbook.

The $70\ \Omega$ and $30\ \Omega$ resistors are in parallel. The equivalent resistance is,

$$70\ \Omega \parallel 30\ \Omega = \frac{70 \times 30}{70 + 30} \\ = 21\ \Omega$$

The $20\ \Omega$ and $5\ \Omega$ resistors are in parallel. The equivalent resistance is,

$$20\ \Omega \parallel 5\ \Omega = \frac{20 \times 5}{20 + 5} \\ = 4\ \Omega$$

Step 2 of 5

Draw the modified circuit diagram.

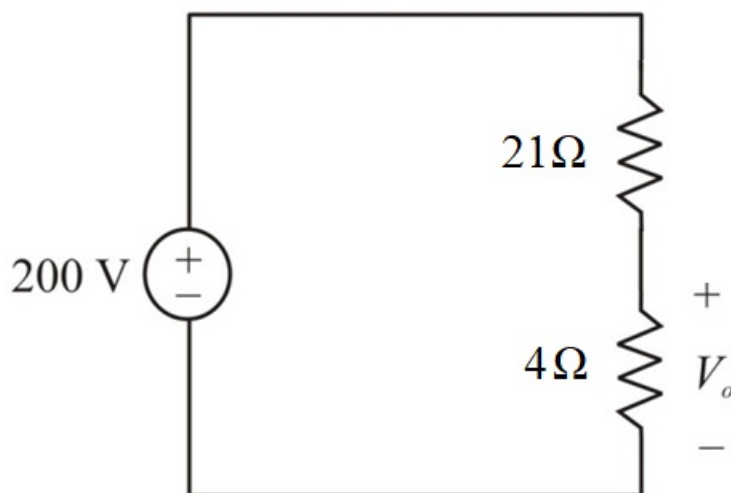


Figure 1

Step 3 of 5

Apply voltage division rule to calculate the output voltage.

$$\begin{aligned}V_o &= \left(\frac{4}{4+21}\right)(200) \\&= \left(\frac{4}{25}\right)(200) \\&= (4)(8) \\&= 32 \text{ V}\end{aligned}$$

Thus, the output voltage is 32 V.

Step 4 of 5

Draw the actual circuit diagram.

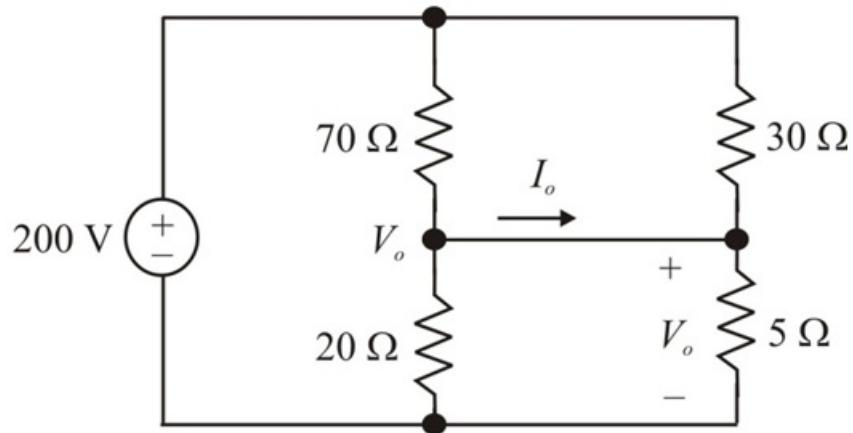


Figure 2

Step 5 of 5

Apply nodal analysis at the node V_o .

$$\frac{V_o - 200}{70} + \frac{V_o}{20} + I_o = 0$$

Substitute 32 V for V_o .

$$\frac{32 - 200}{70} + \frac{32}{20} + I_o = 0$$

$$-2.4 + 1.6 + I_o = 0$$

$$I_o = 0.8 \text{ A}$$

Thus, the output current is 0.8 A.